

UNIT IV

TOOLS AND TECHNOLOGY FOR DISASTER MANAGEMENT

PART A

1	What is early warning system (EWS)? Early Warning Systems (EWS) are integrated processes that identify and forecast potential hazards to provide timely information, allowing individuals, communities, governments and organizations to act and mitigate the impact of disasters. These systems encompass a range of technologies and procedures designed to monitor, predict and communicate imminent risks. The goal is to ensure that populations at risk receive adequate notice and guidance to protect lives, property and the environment.
2	List five components of disaster relief. 1. Search and Rescue Operations. 2. Emergency Medical Assistance. 3. Food, Water and Essential Supplies. 4. Temporary Shelter and Housing. 5. Coordination and Communication.
3	Why is sanitation important in disaster relief efforts? ● Prevents disease outbreaks: Poor sanitation leads to waterborne diseases like cholera and dysentery. ● Preserves dignity and safety: Clean facilities support mental well-being, especially for vulnerable groups.
4	State two reasons for climate change. ● Greenhouse gas emissions: Burning fossil fuels releases CO₂, methane and nitrous oxide. ● Deforestation: Cutting trees reduces carbon absorption and releases stored carbon.
5	What role does GIS play in disaster preparedness and response? ● Risk mapping: Identifies vulnerable zones and critical infrastructure. ● Real-time coordination: Supports emergency response with spatial data and damage tracking.
6	How does information technology contribute to risk assessment in disaster? ● Data integration: Combines satellite, sensor and historical data for predictive modeling. ● Communication systems: Enables early warning alerts and coordination among agencies.
7	What is the purpose of disaster damage assessment? ● Informs response priorities: Identifies areas of greatest need for aid and resources.

	• Supports funding and recovery: Guides reconstruction plans and financial allocations.
8	Mention two elements of climate resilient development. • Adaptation strategies: Infrastructure upgrades, drought-resistant crops, flood defenses. • Mitigation actions: Renewable energy, sustainable land use, emission reductions.
9	What is the significance of standard operating procedures (SOP) in disaster response? • Ensure coordinated actions: SOPs provide clear, step-by-step instructions for various stakeholders (e.g., emergency responders, local authorities, NGOs), reducing confusion during emergencies. • Enhance speed and efficiency: Predefined protocols help streamline response efforts, minimizing delays in lifesaving interventions. • Clarify roles and responsibilities: SOPs assign specific tasks to different agencies or individuals, preventing duplication and gaps in services.
10	Define financial planning in the context of disaster management. Financial planning in disaster management refers to the systematic allocation and management of financial resources to: • Support preparedness, including early warning systems, training and infrastructure. • Fund response operations, such as rescue, relief and rehabilitation. • Enable recovery and reconstruction, including rebuilding homes, infrastructure and livelihoods. • Ensure risk reduction, by investing in mitigation measures (e.g., flood control, earthquake-resistant buildings).
11	What is the role of institutional arrangements in disaster mitigation? Institutional arrangements refer to the structures, frameworks and organizations established to manage disaster risk and implement mitigation measures. Their role includes: • Policy formulation: Developing laws, policies and regulations related to disaster risk reduction. • Coordination: Ensuring collaboration among government agencies, NGOs, private sector and communities. • Capacity building: Training personnel, improving technology and strengthening early warning systems.

	• Monitoring and enforcement: Overseeing implementation of safety standards (e.g., building codes, land-use planning).
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